

Open-source, smart espresso machine

Community Partner: Healthy Buildings Research Lab and COffee And Telesensing Lab

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IP and Compliance Requirements

- Students must sign an NDA: No
- Intellectual Property will be owned by: Negotiable
- Students must be US persons: No
- Other legal requirements (background checks, drug testing, etc): None

Background and Motivation

Brewing espresso is a complex, multiphase process whereby hot, high-pressure water extracts flavorful compounds from roasted coffee beans. To prepare espresso, finely ground coffee is compressed into a tightly packed puck inside a pressure vessel called a portafilter. Water is then forced through the puck at around 6-9 bars of pressure, catalyzing multitudinous physical and chemical reactions. The water dissolves soluble compounds such as acids and sugars, while also emulsifying fats and other complex molecules.

Simultaneously, the flow of hot water releases the coffee's volatile aromatic compounds. Throughout this process, factors such as coffee grind size, water temperature, pressure, and extraction time must be carefully controlled to achieve the desired balance of sweetness, acidity, bitterness, and body in the final espresso "shot."



Figure 1: "Pulling" an espresso shot.

Many newer espresso machines have smart sensors that monitor a variety of factors, such as water temperature, pressure, and extraction time. Features like PID temperature control, pressure profiling, and flow control allow for fine-tuned adjustments to water temperature, pressure, and flow rate, improving consistency and flavor. Additionally, pre-infusion control and timed shot programming further enhance precision. While these advancements provide greater consistency and customization, they come at a higher cost and require a learning curve for operators. The Decent Espresso machine (\$3,699 MSRP), for instance, allows the user to



Figure 2: Decent DE1PRO smart machine. The UI provides real-time information about brewing.

tailor shots by adjusting brew time, pressure, flowrate, and temperature. The Decent also logs shot data and uploads it to the cloud.

Objective

This joint ECE/ME capstone project will design and implement a low-cost, open-source smart espresso machine based on the *Gaggia Classic PRO* platform. Building upon the open-source Gaggiuino project, the team will develop a system capable of precise temperature regulation via PID control, adjustable pump pressure and flow profiling, real-time monitoring, and data logging.

Additionally, the system will include interfaces for laboratory-grade analytical instruments, including Proton Transfer Reaction – Mass Spectrometry (PTR-MS) and UV-VIS spectroscopy. This integration will allow for quantitative analysis of how brewing parameters—such as pressure, temperature, and flow rate—affect the generation and release of aromatic compound and the flavorful compounds that end up in the cup.

The final deliverable will be an open-source, instrumented espresso platform that serves as a research and educational tool for exploring the science of coffee extraction and preparing excellent (rather than decent) espresso.



Figure 3: Gaggia PRO modified with the open-source Gaggiuino hardware to become a smart espresso machine

Requirements

Must:

- 1) The machine must meet or exceed the features of the Gaggiuino Gen 3 STM32/ESP32 project:

<https://gaggiuino.github.io/#/?id=features>

Should:

- 1) Have the ability to “pull a shot” within a controlled environment. That is, the user should be able to prepare a shot in an atmosphere with a set humidity, pressure, and temperature
- 2) Provide interfaces with the PT-RMS in the Healthy Buildings Lab to assess the production of released volatiles

Suggested Skills

This capstone sits at the intersection of electrical engineering, mechanical engineering, and analytical chemistry, so it requires a mix of hardware, software, and scientific skills. Students joining this project should be comfortable with control theory, embedded system design, programming ESP32 and/or STM32, electrical and mechanical computer aided design, and analytical chemistry and instrumentation. Being a coffee nerd helps, and familiarity with the Decent or Meticulous smart espresso machines is a bonus.

Deliverables

Must:

Deliver a machine whose capabilities meet or exceed the features of the Gaggiuino Gen 3 STM32/ESP32 project: <https://gaggiuino.github.io/#/?id=features>